Short questions

|!|BCESA0010000001|!| ★★ SI

Melvin prepared yoghurt by mixing milk and a yoghurt starter culture (plain yoghurt containing living bacteria). The mixture was then incubated at 30 °C for six hours. The pH of the mixture was measured with a pH meter and recorded every hour.

Melvin conducted three trials. The results are recorded as follows:

Time (minute)	pH		
	Trial 1	Trial 2	Trial 3
0	6.5	6.5	6.5
60	6.5	6.4	6.4
120	6.3	6.3	6.2
180	5.8	5.9	5.9
240	5.1	5.2	5.5
300	4.8	4.6	5.8
360	4.5	4.3	4.6

(a) What are the control variables in the above investigation? List ***two***. (2 marks)

_____

_____

(b) What is the purpose of conducting three trials instead of one? (1 mark)

_____

_____

(c) Identify one of the results that appears to be anomalous. (1 mark)

_____

##

(a) Any ***two*** of the following: (2)

- volume / amount of milk
- type of milk
- amount / volume of starter culture
- type of starter culture
- temperature
- time interval

(Or other reasonable answers)

(b) The results are more reliable. / This can reduce the effect of anomalies (or errors) on the results. (1)

(c) Trial 3 at 300 minutes (1)

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(4 marks)

##

	Skills assessed	Marks
(a)	Practical skills	2
(b)	Practical skills	1
(c)	Analysing data	1

|!|BCESB0010000002|!| ★

Read the passage below and answer the questions that follow.

> James Lind (1716–1794) was an English doctor. He carried out experiments to investigate the effect of fresh fruits on curing scurvy (壞血病).
>
> Lind noticed that many sailors suffered from scurvy and their diet usually lacked fresh fruits. He also noticed that once the sailors were back on land and ate fresh fruits again, their scurvy disappeared.
>
> In his experiment, Lind chose 20 sailors suffering from scurvy and put them into two groups, I and II. He gave group I a diet including fresh fruits, and gave group II the same diet but without fresh fruits. Group I recovered from scurvy, while group II did not.

Identify the following aspects of Lind's investigation:

(a) Hypothesis (1 mark)

(b) The control group (1 mark)

(c) Observation (1 mark)

(d) Conclusion (1 mark)

##

(a) A lack of fresh fruits may cause scurvy. (1)

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(b) The sailors whose diet was without fresh fruits (1)

(c) Many sailors suffered from scurvy and their diet usually lacked fresh fruits. Once the sailors were back on land and ate fresh fruits again, their scurvy disappeared. (1)

(d) Eating fresh fruits cures scurvy. (1)

(4 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Analysing data	1
(c)	Analysing data	1
(d)	Analysing data	1

|!|BCESB0010000003|!| ★

Scientific investigations should be carefully designed, so that the data obtained is accurate, and the results are valid and reliable. For each of the terms listed in column 1, select from column 2 a suitable meaning. Put the appropriate letter in the space provided.

(3 marks)

Column 1		***Column 2***
Accurate	(a) _____	A. conduct repeated trials to obtain similar results
Valid	(b) _____	B. choose appropriate apparatus for taking measurements
Reliable	(c) _____	C. obtain measurements that are closest to the true values

##

(a) C (1)

(b) B (1)

(c) A (1)

(3 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	1
(b)	Applying concepts	1
(c)	Applying concepts	1

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|!|BCESB0010000004|!| ★ SI

Grandparents may advise young children not to take a cold shower or they may catch a cold. Peter wants to test scientifically whether this belief is correct or not. He designed an investigation as follows:

1. Recruit 20 volunteers and divide them into two groups.
2. One group takes a cold shower every day, while the other group takes a hot shower every day for a month.
3. Record the number of volunteers who catch a cold during the period.

(a) Is the advice 'do not take a cold shower or you may catch a cold' a conclusion or hearsay? (1 mark)

_____

(b) What is the hypothesis of Peter's investigation? (1 mark)

_____

_____

(c) What are the control variables in the investigation? List ***two***. (2 marks)

_____

_____

(d) Suggest and explain ***one*** way to improve the reliability of Peter's investigation. (2 marks)

_____

_____

##

(a) hearsay (1)

(b) Taking a cold shower may increase the chance of catching a cold. (1)

(c) Any ***two*** of the following: (2)

- age and health conditions of the volunteers
- diet
- level of physical activity
- temperature of the cold shower
- duration of the cold shower

(Or other reasonable answers)

(d) Recruit more volunteers (e.g. 100) to increase the sample size. (1)

This can reduce the effect of individual differences on the results. (1)

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(6 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	1
(b)	Analysing data	1
(c)	Applying concepts	2
(d)	Applying concepts	2

|!|BCESB0010000005|!| ★

Respiration and excretion are two of the characteristics of living things.

(a) State ***three*** other characteristics of living things. (3 marks)

(b) Explain the difference between respiration and breathing. (2 marks)

(c) Suggest why excretion is important to living organisms. (2 marks)

##

(a) Any ***three*** of the following: (3)
nutrition / movement / sensitivity / growth / reproduction

(b) Respiration is the breakdown of food molecules in cells to release energy (synthesize ATP molecules). (1)
Breathing refers to the moving of air into and out of the lungs. (1)

(c) Excretion refers to the removal of metabolic wastes from the body. (1)
This can avoid the accumulation of wastes inside the body, which may lead to poisoning. (1)

(7 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	3

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(b)	Applying concepts	2
(c)	Applying concepts	2

|!|BCESB0010000006|!| ★

Certain non-living things, such as an air-conditioner, show some characteristics similar to those of living things.

(a) State which characteristic of organisms is comparable to each of the features of an air-conditioner.

(i) obtaining energy from an alternating current power supply (1 mark)

_____

(ii) release of waste gases (1 mark)

_____

(iii) switching off when the room temperature reaches a set point (1 mark)

_____

(b) State ***two*** characteristics of organisms that are ***not*** displayed in an air-conditioner. (2 marks)

_____

_____

##

(a) (i) nutrition (1)

(ii) excretion (1)

(iii) sensitivity (1)

(b) Any ***two*** of the following: (2)

respiration / movement / growth / reproduction

(5 marks)

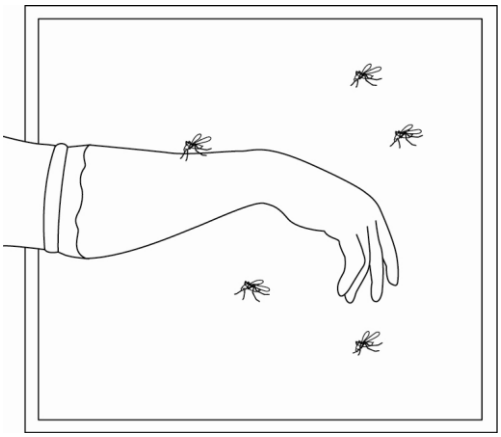
##

	Skills assessed	Marks
(a)(i)	Applying concepts	1
(a)(ii)	Applying concepts	1
(a)(iii)	Applying concepts	1
(b)	Understanding basic concepts	2

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|!|BCESA0010000007|!| ★★ **SI**

Scientists carry out an experiment to investigate the effectiveness of a new mosquito repellent (chemical X). They perform the experiment by spraying the chemical X onto the arms of a number of people. These people then put their arms into a sealed tank containing mosquitoes, for 15 minutes. The following diagram shows the set-up.



The number of mosquito bites on each arm of five people was counted and compared with a control group. The following table shows the results.

Person	**Number of mosquito bite**	
	Chemical X	**Control**
A	1	13
B	0	18
C	1	12
D	2	13
E	1	14
Average	1	

(a) Calculate the average number of bites for the control group. (1 mark)

(b) In the control group, the people also put their arms into a sealed tank containing mosquitoes, but how is the control set-up different from the experimental set-up? (1 mark)

(c) Suggest ***two*** factors that need to be controlled in this experiment. (2 marks)

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(d) What conclusion can be drawn from the results of the experiment? (1 mark)

_____

_____

(e) State ***one*** advantage of the use of mosquito repellent in public health. (1 mark)

_____

##

(a) 14 (1)

(b) Any ***one*** of the following: (1)

- Their arms are sprayed with water.
- Their arms are not sprayed with chemical X.

(Or other reasonable answers)

(c) Any ***two*** of the following: (2)

- number of mosquitoes
- species of mosquitoes
- temperature

(Or other reasonable answers)

(d) Chemical X can reduce the number of bites by mosquitoes. (1)

(e) It can prevent the transmission of disease transmitted by mosquitoes (e.g. dengue fever.) (1)

(6 marks)

##

	Skills assessed	Marks
(a)	Mathematical skills	1
(b)	Practical skills	1
(c)	Practical skills	2
(d)	Analysing data	1
(e)	Applying concepts	1

|!|BCESC0010000008|!|

Complete the following paragraph with suitable words. (5 marks)

(a)_____ is the study of life and living organisms. Scientists are curious about unexplained events and ask questions about their (b)_____. They then make a proposed explanation called (c)_____. After this, they design and perform (d)_____. If the results support what they think, a (e)_____ can be drawn.

##

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(a) Biology (1)
(b) observations (1)
(c) hypothesis (1)
(d) experiments (1)
(e) conclusion (1)

(5 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1
(e)	Understanding basic concepts	1

|!|BCESC0010000009|!|
For each of the branch of biology listed in column 1, select from column 2 one example of the study that belongs to that branch. Put the appropriate letter in the space provided.

(4 marks)

Column 1		***Column 2***
Ecology	(a) _____	A. inheritance of characteristics
Cytology	(b) _____	B. mutualism
Evolutionary biology	(c) _____	C. natural selection
Genetics	(d) _____	D. nuclear division

##

(a) B (1)
(b) D (1)
(c) C (1)
(d) A (1)

(4 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	1
(b)	Applying concepts	1
(c)	Applying concepts	1
(d)	Applying concepts	1

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|!|BCESC0010000010|!|

For each of the common characteristics of organisms listed in column 1, select from column 2 one phrase that matches it. Put the appropriate letter in the space provided.

(3 marks)

Column 1		***Column 2***
Excretion	(a) _____	A. the metabolic process that breaks down food molecules in cells to release energy
Respiration	(b) _____	B. the ability to detect changes in the environment and make responses
Sensitivity	(c) _____	C. the increase in body size and complexity
		D. the removal of the wastes produced from chemical reactions inside the body

##

(a) D (1)

(b) A (1)

(c) B (1)

_____

(3 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1

|!|BCESB0010000011|!| ★

Figure A and **Figure B** represent two of the seven common characteristics of living organisms.

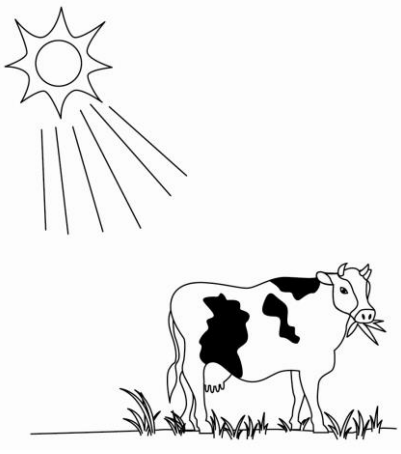


Figure A

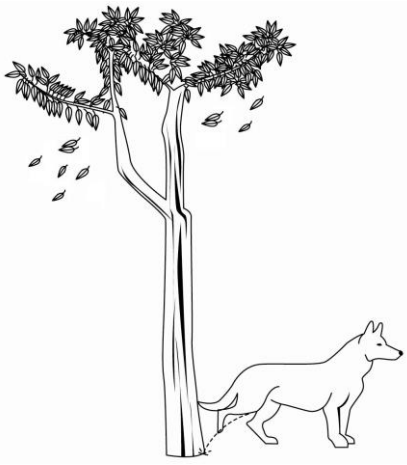


Figure B

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(a) Identify the characteristic shown in **Figure A**. Explain your answer. (3 marks)

(b) Identify the characteristic shown in **Figure B**. Explain your answer. (3 marks)

##

(a) Figure A shows nutrition. (1)
The grasses carry out photosynthesis under sunlight to produce their own food. (1)
The cow eats the grasses as food. (1)
(b) Figure B shows excretion. (1)
The tree stores waste products in leaves which then fall off. (1)
The dog excretes wastes in urine. (1)

(6 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
	Applying concepts	2
(b)	Analysing data	1
	Applying concepts	2

|!|BCESB0010000012|!| ★

A is a living thing. B, although exhibiting some of the features observed in A, is regarded as a non-living thing.



A



B

(a) State ***one*** common characteristic of living things that is demonstrated by both A and B. (1 mark)

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(b) State ***three*** differences between A and B in terms of the seven common characteristics of living organisms. (3 marks)

##

(a) Movement (1)

(b) Any ***three*** of the following comparisons: (3)

A	B	
- consumes food for energy	- requires the combustion of fuels as energy	(1)
- can grow until it reaches a certain age	- the size is fixed and cannot increase	(1)
- can reproduce to give birth to offspring	- unable to reproduce	(1)
- can detect stimuli in the environment and respond to them actively	- can detect some external stimuli but often cannot make active responses to the stimuli	(1)

(4 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Applying concepts	3

|!|BCESB0010000013|!| ★

The following photograph shows a cheetah.



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(a) Suggest ***one*** common characteristic of life that can be observed in the photograph. (1 mark)

(b) Briefly explain the significance of the characteristic mentioned in (a). (2 marks)

##

(a) Movement (1)

(b) It allows the cheetah to search for food. (1)

It allows the cheetah to find mates. (1)

(Or other reasonable answers)

(3 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Applying concepts	2

|!|BCESB0010000014|!| ★

Ann found two fish of the same species in an aquarium. She tried to catch the two fish using a net but they moved away immediately. After one month, one more fish of the same species appeared in the aquarium.

(a) According to the above descriptions, name ***three*** common characteristics of living organisms that the fish showed. (3 marks)

(b) Explain the significance of the three characteristics to the fish. (3 marks)

##

(a) sensitivity (1)

movement (1)

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reproduction (1)

(b) The fish can detect the presence of predators and escape. / The fish can detect the presence of food and eat it. (1)

The fish can move away from the dangers in the environment. (1)

Reproduction ensures that the species does not die out. (1)

(6 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	3
(b)	Applying concepts	3

|!|BCESB0010000015|!| ★ **SI**

A student observed that plants grow well in sunny areas and grow poorly in shady areas. He made a hypothesis and then carried out an experiment to test whether light is required for plants to grow well.

(a) State the hypothesis made by the student. (1 mark)

(b) Outline how you would carry out an experiment to test the hypothesis in (a). (3 marks)

(c) State ***one*** control variable in this experiment. (1 mark)

##

(a) Light is necessary for the growth of plants. (1)

(b) Place three seedlings into a conical flask containing nutrient solution. (1)

Place the conical flask under bright light for a week. (1)

Set up a control by placing the same set-up in darkness (e.g. inside a cupboard). (1)

(Or other reasonable answers)

(c) Any ***one*** of the following: (1)

species of the plants / age of the plants / volume of the nutrient solution / concentration of the nutrient solution

(Or other reasonable answers)

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_____

(5 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	1
(b)	Applying concepts	3
(c)	Applying concepts	1

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